Speech-like Analysis of Snore Signals for the Detection of Obstructive Sleep Apnea

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Abstract – **Correlation between snoring and Obstructive Sleep Apnea (OSA) has been studied over the past years using techniques such as sound intensity computation, power spectrum estimation, counting of snores, and more recently, pitch and jitter analysis. Although these studies have ascertained the relationship between snoring and OSA, they gave mixed outcomes in diagnosing OSA.**

In this paper, snore signals are analyzed using Linear Predictive Coding (LPC), which is a powerful speech analytic tool. The noisy snore signals were first preprocessed using a Modified Normalized Least-Mean-Square (MNLMS) adaptive filter for noise cancellation, followed by the extraction of formant structures from the denoised snore signals to discriminate between benign and apneic snores.

This initial study demonstrates that the formant structures of snore signals contain useful information for OSA detection, and suggests the feasibility of using snore signal analysis as a non-invasive and inexpensive diagnostic tool for mass screening of OSA.

Keywords – **Obstructive sleep apnea, Snoring, Normalised least-mean-square adaptive filter, Linear predictive coding, Formant**

I. INTRODUCTION

Obstructive Sleep Apnea (OSA) is a common sleep-related breathing disorder. A study done by Young et al [1] on 602 state employees have found that the incidence of OSA was about 24% in men and 9% in women. However, many researchers believe that up to 82% of men and 93% of women with moderate to severe OSA remain undiagnosed [2]. In Singapore, OSA affects about 15% of the total population [3].

OSA is defined as repetitive pauses in respiration for at least 10 s, corresponding to a complete occlusion of the upper airway, despite continuous chest and abdominal movements, and accompanied by chronic snoring, a decrease in oxygen saturation and an arousal from sleep. Therefore, the common symptoms of OSA include snoring, choking at night, and frequent arousals. Daytime manifestations are fatigue, excessive daytime somnolence, poor concentration, and irritability. Untreated OSA may lead to many life-threatening complications, such as coronary artery disease, cerebral vascular accidents, and congestive heart failure [1-3].

Polysomnography (PSG), performed over a full night sleep, is currently the gold standard diagnostic tool for assessing OSA. In the PSG assessment, numerous physiologic electrodes are placed on the patient in order to record brain electrical activity, eye and jaw muscle movement, leg muscle movement, airflow, respiratory effect, and oxygen saturation. Therefore, the high cost of the in-sleep-center diagnosis, the limited number of available sleep laboratories, and the discomfort of the electrodes attached to the head and body of the patients, are possible limitations of PSG. Hence, it would be desirable to have an alternative method to diagnose OSA non-invasively, with greater comfort and at a lower cost.

Snoring, a major diagnostic marker for OSA, is affecting about 40-60% of adults [4]. It is resulted from the vibration of soft tissues of the upper airway, which in turn is caused by turbulent airflow created by narrowing of cross-sectional area in the upper airway. A mathematical model [5] of snoring has been developed in terms of airflow through an elastic tube. When the geometry, elasticity, resistance, and flow are altered correctly, the tube can be made to open and close repetitively, leading to production of sound that resembles snoring. If the parameters are altered further, a complete collapse of the tube can be created, simulating OSA. This model suggests that OSA attacks are primarily caused by anatomical and functional abnormalities in the upper airway. As such, any structural changes in the upper airway, which acts as a variable acoustic filter, are reflected in the acoustical properties of the snore signals, just like the vocal tract in the event of speech signals.

Several studies have reviewed the relationship between snoring and OSA using methods such as counting number of snores [6], employing variance changes [7] and snoring sounds intensity [8], estimating power spectrum [9,10], and analyzing pitch and jitter [11,12] in snore signals. However, most of these methods may not appropriately reveal the essential acoustic properties of the upper airway. Another important consideration is the presence of noise. Even in the well-controlled sleep laboratory, the recorded snoring sounds are highly corrupted by background acoustical noise and

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electromagnetic interference (EMI), which in turn, can seriously affect the accuracy of analysis.

Linear Predictive Coding (LPC) is a well-recognized speech analysis technique that can be used to derive appropriate vocal tract frequency models for speech signals [13]. Since the production of speech and snoring sounds share many similarities as described above, it is also reasonable to assume that snore signals could be well modeled using LPC. The resonances of the upper airway are known as formants, which are manifestations of energy maxima at the resonant frequencies. The first three formants (F1, F2, and F3) may carry important cues in the characterization of snore signals, since they are directly related to the acoustic analogues of the upper airway. Studies in speech analysis have shown that F1 and F2 relate to the location and amount of constriction in the upper airway, while F3 corresponds to the length of the upper airway [14].

This study takes the above considerations into account. First, noise embedded in the raw snore signals are removed using a Modified Normalized Least-Mean-Square (MNLMS) adaptive filter. Next, using the noise-suppressed snore signals, the Linear Predictive Coding (LPC) method is implemented to review information on the acoustical structure of the upper airway by estimating its formant frequencies. Lastly, the formant frequencies are used to classified between benign (apnea index AI < 5 events/h) and apneic (AI ≥ 5 events/h) snore signals, from actual clinical PSG cases.

II. Materials and Methods

A. Snore Acquisition System

Fig. 1 shows a high fidelity snore acquisition system that was built in a sleep laboratory in the Sleep Disorders Unit, Singapore General Hospital. Two non-invasive unidirectional condenser microphones (20-20000 Hz, SM81, Shure Incorporated) were used for snore signals recording; one placed approximately 30 cm above the mouth of the patient, while the other placed near the air-conditioner which contributed most of the acoustical noise. The snore signals were amplified by low-noise pre-amplifiers (20-22000 Hz, FP23, Shure Incorporated), transmitted through a double shielded coaxial cable to reduce EMI, and captured by a dynamic signal acquisition card (NI4552, National Instruments). The sampling rate was set at 44.1 kHz in order to capture high quality snore signals and any speech integrated within the snore signals.

Snoring sounds have a wide dynamic range of more than 90 dB; therefore, low intensity snore signals can be easily masked by noise source. Furthermore, background acoustical noise, for example, humming sound from the air-conditioner, footstep sound along the laboratory's corridor, and EMI resulted from nearby PSG machine and power lines, play a significant role in decreasing the signal-to-noise ratio (SNR) of the acquired snore signals, leading to poor accuracy. Thus, in this work, SNR enhancement was attained through careful hardware based snore acquisition system design as illustrated in Fig, 1, and through software based MNLMS adaptive filter design.

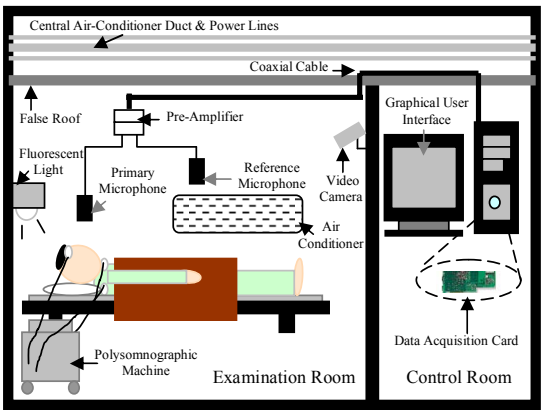


Figure 1. Setup of snore acquisition system

B. Human Subjects & Protocols

Eight apneic patients (aged 41 ± 12, body mass index BMI = 34 ± 9 kg/m^2, AI = 44 ± 28 events/h), and five benign patients (aged 42 ± 11, BMI = 26 ± 4 kg/m^2, AI = 1 ± 1 events/h) were studied.

10 episodes of snoring were randomly selected from each patient for analysis, giving a total of 130 episodes. For apneic patients, snoring corresponding to the 1st, 2nd and 3rd breath after an OSA attack, was included. The snores were then grouped two different classes: benign snore and apneic snore, by trained clinicians in the sleep disorder unit.

C. Proposed Normalized Least-Mean-Square Adaptive Filter

An adaptive noise canceller [15] attempts to cancel out the stochastic noise present in the primary input (primary microphone), by maximal matching of a noise reference input (reference microphone). The primary input $x(n)$ is assumed to be consisted of the clean snore signal $s(n)$ corrupted by additive noise $v_1(n)$; whereas the reference input $\mathbf{v}(n)$ is assumed to be uncorrelated with $s(n)$, but correlated with $v_1(n)$ (see Fig. 2).

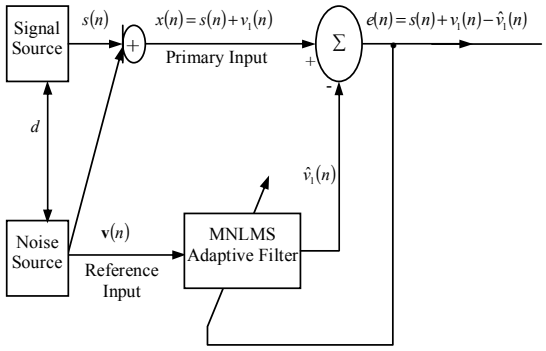


Figure 2. Adaptive noise canceller

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The aim of an adaptive filter is to minimize the output noise without distorting the signal of interest, thereby increasing the SNR at the output. This achievement can be accomplished by proper selection of its tap-weights vector **w**. In this paper, the tap-weights **w** are adapted by means of a MNLMS algorithm, which obeys the principle of minimal disturbance, while incorporating a delay d in $v_1(n)$ relative to $\mathbf{v}(n)$, to account for the distance between the two microphones as illustrated in Fig. 2.

Hence, the corresponding algorithm is modified as

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu}{\varepsilon + \|\mathbf{v}(n-d)\|^2}\mathbf{v}(n-d)e(n) \tag{1}$$

where μ is the dimensionless step-size, ε is a small positive constant (used to avoid a division-by-zero error when $\|\mathbf{v}(n\text{-}d)\|^2$ is very small), $e(n)$ is the system output, and

$$\|\mathbf{v}(n-d)\|^2 = \sum_{m=0}^{M-1}|\mathbf{v}(n-d-m)|^2 \tag{2}$$

is the total instantaneous energy of the input signal $\mathbf{v}(n\text{-}d)$, estimated over values within the tapped-delay line. The delay parameter d may be determined by performing cross-correlation between the primary and the reference signals. The MNLMS adaptive filter has all the favorable benefits of the conventional LMS and NLMS adaptive filters with the possible additions of faster convergence and higher stability.

D. Snore-Formant Structures

Linear Predictive Coding (LPC) is a parametric method of analysis in which parameters are defined for a mathematical model describing the signals. Fig 3 illustrates a p-order LPC model for the production of snore signals. The excitation source $u(n)$ is modulated by the gain factor G, and the scaled source is used as an input to the upper airway. The upper airway acts as an autoregressive acoustic filter $H(z)$ whose coefficients $a_1, a_2, \ldots, a_p$ closely approximate the reality of snoring sounds produced by the patients.

In this work, the predictor coefficients are computed using the auto-correlation method [16] programmed in MATLAB™. This method guarantees all poles to be inside the unit circle, and thence, yielding stability in $H(z)$. The Levinson-Durbin recursion (3) is utilized to compute the coefficients for the auto-correlation method by solving the Yule-Walker equations. Formants are identified by peak-picking the LPC spectrum.

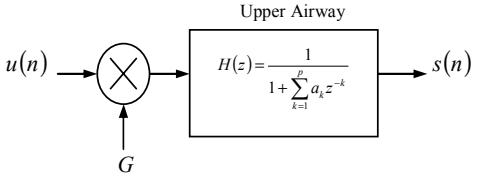


Figure 3. Snore signals production model

$$E_0 = R(0) \tag{3a}$$

$$k_i = -\left[R(i) + \sum_{j=1}^{i-1} a_j^{(i-1)} R(i-j)\right] \Big/ E_{i-1} \quad , \quad 1 \le i \le p \tag{3b}$$

$$a_i^{(i)} = k_i \tag{3c}$$

$$a_j^{(i)} = a_j^{(i-1)} + k_i a_{i-j}^{(i-1)} \quad , \quad 1 \le j \le i-1 \tag{3d}$$

$$E_i = (1 - k_i^2)E_{i-1} \tag{3e}$$

where $R(i)$ is the autocorrelation of the process, E_i is the minimum total squared error, k_i denotes the intermediate quantities, and a_j is the predictor coefficients. Equations (3a)-(3e) are solved recursively for $i = 1, 2, \ldots, p$, thereby giving the final solutions as

$$a_j = a_j^{(p)} \quad , \quad 1 \le j \le p\,. \tag{3f}$$

III. Results and Discussion

For optimum performance of adaptive noise cancellation, the following parameters were selected: $\mu = 0.1$, $\varepsilon = 10^{-8}$, and $d = 220$. The value for step-size μ was chosen as a compromise between small excess mean-square error and fast convergence rate; whereas, the value for sample delay d was obtained by performing the cross-correlation function between the primary and reference signals, as shown in Fig. 4. Alternatively, the value for sample delay d may be determined by using the acoustic propagation time properties, i.e. measuring the distance between the primary and reference microphone, calculating the acoustic propagation time, and converting the time delay to digital sample delay.

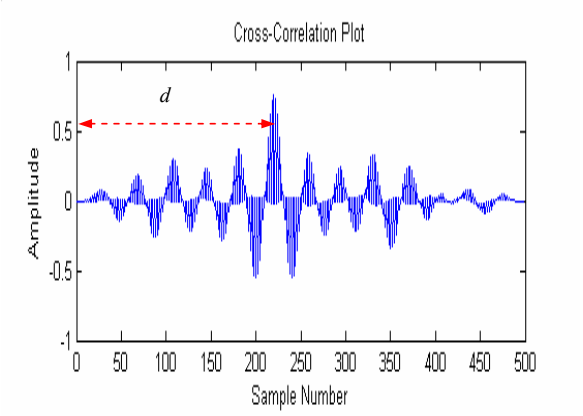


Figure 4. Cross-correlation function for sample delay estimation

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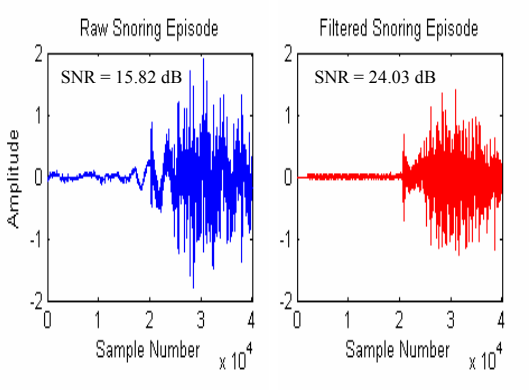


Figure 5. Noise cancellation using MNLMS adaptive filter

Fig. 5 exhibits the results of the MNLMS adaptive noise cancellation. The left plot shows a raw snoring episode with an initial SNR = 15.82 dB, and the right plot shows the same signals after the noise cancellation yielded a SNR = 24.03 dB, an increment of SNR = 8.21 dB. Generally, the MNLMS adaptive filter could achieve a SNR improvement from 6 to 9 dB. Improvements by the MNLMS adaptive filter were also confirmed by listening tests performed by a group of skilled clinicians in the sleep laboratory.

Fig. 6 illustrates the formant structures of common benign and apneic snores in a 14th-order LPC spectrum. The choice of p = 14 was arrived at through experimentation. It was found that, with lesser predictor coefficients, the merging formants of certain snore signals were represented by only one complex pole-pair, which is undesirable.

Based on the existing dataset of 130 denoised episodes of snoring, the formant frequencies range of both benign and apneic snores are tabulated in Table I. The quantitative differences in the formant frequencies may be explained in terms of the differences in upper airway anatomy or mechanics in patients with and without OSA. Patients with OSA usually have weak muscle tone, and unstable upper airway that collapses easily.

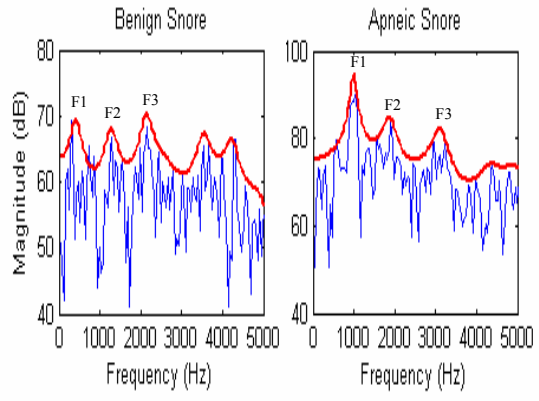


Figure 6. Formant structures of a common benign and apneic snore

TABLE I. A RANGE OF FORMANT (KHZ) FOR BENIGN AND APENIC SNORES

Type of Snore	F1	F2	F3
Benign	0.3 – 0.8	1.4 – 1.9	2.4 – 3.0
Apneic	0.6 – 1.3	1.7 – 2.3	2.6 – 3.5

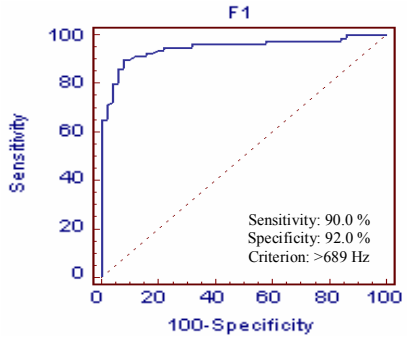


Figure 7. Receiver operating characteristic curve

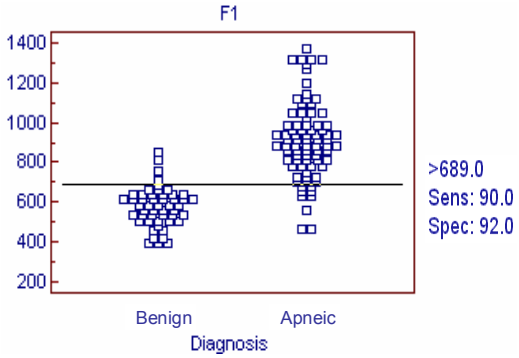


Figure 8. Interactive dot diagram

The diagnostic performance of this approach was further evaluated using Receiver Operating Characteristic (ROC) curve analysis and interactive dot diagram [17], as shown in Fig. 7 and 8 respectively. The results obtained are: sensitivity = 90%, specificity = 92%, and criterion value > 689 Hz.

IV. CONCLUSIONS

The purpose of this work is to study the feasibility of using acoustical signatures in snore signals for non-invasive diagnosis of OSA. A Modified Normalized Least-Mean-Square (MNLMS) adaptive filter is implemented for noise cancellation, bearing an SNR improvement from 6 to 9 dB. With the efficacious cancellation of noise and consideration of the delay in signals between the two snore acquisition microphones, the formant structures of snore signals were found to be indicative of benign and apneic snores, with a sensitivity of 90%, and specificity of 92%; and that a decision

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boundary of F1 ≈ 690 Hz best distinguish between benign patients and OSA patients. Thus, the present results also supported the hypothesis that the snore signals of benign and apneic patients may have distinguishable acoustic features.

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